Confidently Wrong: Silent Memory Failures in LLMs Under Interference

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Abstract

Across 39 large language models, we identify a silent memory failure under in-context interference: models confidently return wrong values with fewer than 1% hallucination. The silence has structure: RI errors are fully silent (0% refusal), while PI errors carry a weak uncertainty signal ($\sim$4% refusal)—an asymmetry in when models flag a failure. We adapt the classical AB-AC paired-associate interference paradigm (Barnes and Underwood, 1959)—46 category-value pairs followed by N interleaved updates per category ($N \in \{3, 10, 50, 100, 200, 300\}$)—to test retrieval of either the initial value (retroactive interference, RI) or the most recent value (proactive interference, PI) on identical stimulus sequences. Every model exhibits the same pattern: PI causes substantially more degradation than RI (Cohen's $d = 1.73$, $p < 0.0001$), inverting the typical human pattern. The failure has additional structure: model size predicts RI resistance ($R^2 = 0.49$) but not PI resistance ($R^2 = 0.06$, n.s.), and chain-of-thought reasoning models improve RI while worsening PI. Error analysis reveals distinct failure modes: RI failures are passive retrieval failures (51%), while PI failures show active primacy intrusion (56%). These results characterize a structured epistemic blind spot: LLMs have partial self-knowledge about their memory failures, asymmetric between the two interference types, with implications for reliability in interference-heavy deployment settings.

1. Introduction

Consider a medical AI tracking a patient's blood pressure across a 12-hour emergency visit: 120 mmHg at triage, 135 after 15 minutes, 128 post-medication, 118 at discharge. When a physician later asks for the initial value, the system returns 128—and gives no indication it's wrong. No refusal, no "I'm uncertain," no flag. This is a silent retrieval failure: the model produces a confident wrong answer, and neither the model nor the user can detect it from the output alone.

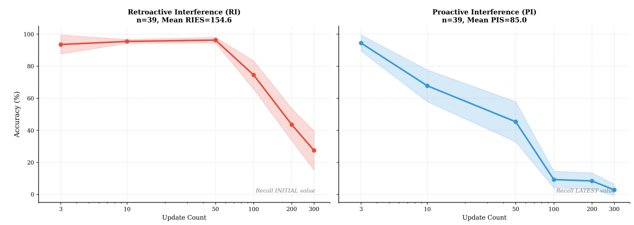

Figure 1. **LLMs Show Opposite Interference Pattern to Humans.** Retroactive interference (RI, left) maintains near-ceiling accuracy through N=50 (96%), then drops sharply to 25% at N=300. Proactive interference (PI, right) collapses accuracy from 94% to 3%—near-complete failure. All 39 models show PI > RI, contrasting with human immediate recall, where recency dominates. Shaded regions: 95% CI.

We show this failure is universal, structured, and invisible. Testing 39 LLMs of diverse architectures and scales, we adapt the classical AB-AC paired-associate interference paradigm (Barnes and Underwood, 1959) to probe two retrieval tasks on identical stimulus sequences: recalling the initial value of a key after N interleaved updates (*retroactive interference*, RI) or recalling the latest value (*proactive interference*, PI). Every model tested shows the same pattern: PI causes substantially more accuracy loss than RI (Cohen's $d = 1.73$, $p < 0.0001$), inverting the typical human finding where recency dominates recall (Murdock, 1962; MacLeod, 2024). Across both conditions, fewer than 1% of errors are hallucinations; refusals are 0% for RI and $\sim$4% for PI—an asymmetry in when models produce any uncertainty signal at all (Figure 1).

The failure is structured, not random. Model size predicts resistance to RI ($R^2 = 0.49$) but not to PI ($R^2 = 0.06$, n.s.)—scaling parameters helps one failure mode but not the other. Chain-of-thought reasoning models (Wei et al., 2022; OpenAI, 2024) follow the same dissociation in reverse: they

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improve RI at the cost of PI. And the failure modes differ qualitatively: RI failures are passive (the model cannot locate the initial value); PI failures show active primacy intrusion (the model returns an earlier value in place of the latest). These dissociations suggest RI and PI involve functionally distinct processes; we return to the mechanism question in the Discussion.

Prior LLM memory research has focused on PI alone (Wang and Sun, 2025), leaving RI largely unexplored. Yet applications from medical records to legal documents to agentic workflows require recalling initial states after updates—and do so under silent failure conditions where the absence of an uncertainty signal is itself a reliability concern.

Contributions.

1. **A silent failure with structured asymmetry**: Across 39 LLMs, in-context interference produces confident wrong answers with fewer than 1% hallucination. RI failures are fully silent (0% refusal); PI failures are partially signaled ($\sim$4% refusal).
2. **Structured dissociation**: Model size predicts RI resistance ($R^2 = 0.49$) but not PI ($R^2 = 0.06$); reasoning models invert the trade-off.
3. **Error taxonomy**: RI failures are passive (retrieval failure, 51%); PI failures are active (primacy intrusion, 56%)—position confusion, not fabrication.

2. Related Work

We study *in-context interference*—competition between items within a single forward pass—distinct from *catastrophic forgetting*, where new training overwrites prior knowledge in model weights (McCloskey and Cohen, 1989; French, 1999). Existing long-context benchmarks primarily test positional search: whether a model can locate specified information within a long input (Kamradt, 2023; Bai et al., 2024; Modarressi et al., 2025; Fu et al., 2025; Ling et al., 2025). These tasks do not manipulate semantic competition between items and therefore do not probe retrieval under interference.

Transformer attention exhibits known positional biases—models struggle with middle positions ("lost in the middle") (Liu et al., 2024) and disproportionately attend to initial tokens ("attention sinks") (Xiao et al., 2024). How these biases interact with in-context interference is untested; we engage this alternative account in the Discussion.

The silent nature of our observed failure connects to a broader literature on LLM epistemic limits. Kadavath et al. (2022) show that LLMs have *partial* self-knowledge—they can often predict whether they will answer a question correctly—while Yin et al. (2023) show they fail to recognize questions they cannot answer. Our results identify a complementary regime: in-context retrieval under semantic interference, where more than 97% of wrong retrievals produce no uncertainty signal, with asymmetric refusal rates across RI and PI. The closest prior LLM memory result is Wang and Sun (2025), who finds that model size predicts PI resistance ($R^2 = 0.26$) while context length does not; retroactive interference and its relationship to PI remained unstudied until this work.

3. Methods

Task Design. We adapt the classic AB-AC paired-associate interference paradigm (Barnes and Underwood, 1959) to probe LLM memory. Models track 46 category-value pairs (e.g., `visual art: impressionism`), then process N interleaved updates per category ($N \in \{3, 10, 50, 100, 200, 300, 400\}$), where each update assigns a new value to a previously seen category (e.g., `visual art: baroque`). We then query either the initial value (retroactive interference, RI) or the most recent value (proactive interference, PI). Critically, RI and PI use identical stimulus sequences—only the query target differs—enabling controlled comparison (Figure 2). All experiments use temperature $= 0$ for deterministic output; session independence derives from fresh random sequences, not sampling variance (details and adaptive-stopping rule in Appendix B).

Dataset. We constructed a dataset of 46 semantic categories with 500–750 real-world named values per category (e.g., "retinal art," "roofer's hatchet"). Updates are randomly interleaved across categories without grouping updates for the same key contiguously—mimicking concurrent updates in real-world data logs. This setup ensures interference arises from semantic competition, not positional proximity. Details in Appendix A.

Metrics. To capture overall interference resistance (not just performance at one level), we compute the **Retroactive Interference Endurance Score (RIES)**—the area under the accuracy-vs-interference curve, log-scaled to weight performance equally across interference magnitudes:

$$\text{RIES} = \int_{N=3}^{300} A_{\text{RI}}(N)\, d(\log_{10}(N+1)) \tag{1}$$

where $A_{\text{RI}}(N)$ is retrieval accuracy for initial values at interference level N. The log-transformed integration range spans $\log_{10}(4)$ to $\log_{10}(301)$, giving a width of ≈ 1.88; consequently, RIES $\approx 1.88 \times \bar{A}_{\text{RI}}$, where $\bar{A}_{\text{RI}}$ is mean accuracy across interference levels. A model scoring 100% accuracy at all levels achieves RIES ≈ 188; a model at 50% throughout scores ≈ 94. **PIES** (Proactive Interference Endurance Score) uses the identical formula applied to accuracy for most-recent values. Higher scores indicate

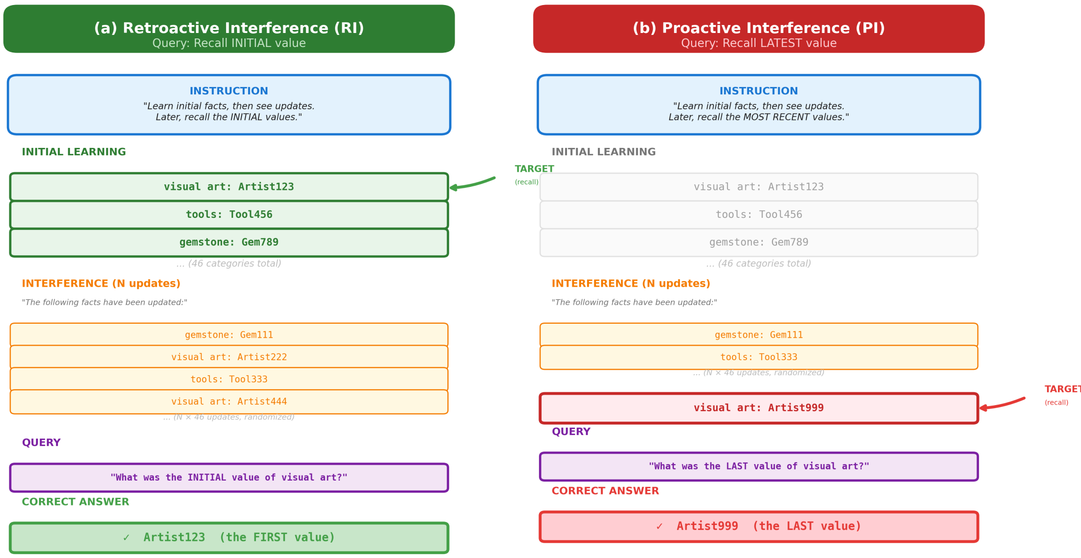

Figure 2. **Experimental Paradigm.** Both RI and PI use identical sequences: 46 initial facts followed by N updates. RI queries initial values (green); PI queries final values (red). This isolates interference from search difficulty.

stronger resistance to interference. Thus, RIES > PIES means a model resists RI better than PI—equivalently, PI causes more performance degradation than RI.

Models. We tested 39 LLMs with complete data across all interference levels, spanning approximately 1B to 2.5T parameters. For proprietary models where exact parameter counts are not publicly disclosed, we use publicly-reported estimates; full specifications in Appendix C. Models include open-weight (Llama, Qwen, DeepSeek) and proprietary (GPT-5, Claude-4.5, Gemini, o1, o3) architectures, with context windows from 8K to 2M tokens.

4. Results

4.1. Baseline Decay: PI Collapses Faster than RI

All 39 tested models exhibit substantial interference under both conditions. For **retroactive interference (RI)**, mean accuracy remains near ceiling through $N = 50$ (96.0%), then drops sharply: 72.9% at $N = 100$, 45.5% at $N = 200$, and 24.6% at $N = 300$—a $\sim 4\times$ decline from peak.

Proactive interference (PI) is far more severe. Mean accuracy starts at 94% at low interference but collapses to 3% at $N = 300$—near-complete failure. The steepest decline occurs between $N = 10$ and $N = 100$, with no ceiling plateau (Figure 1).

PI > RI is universal. Every model tested (39/39) shows higher RIES than PIES; the mean RIES–PIES gap is 69.5 points (paired $t(38) = 10.71$, $p < 0.0001$, Cohen's $d = 1.73$). This inverts the typical human pattern, where recency dominates immediate recall (Murdock, 1962; MacLeod, 2024).

Performance varies substantially across models (Table 1). Top performers (o1, o3) maintain near-perfect accuracy even at maximum interference, while bottom performers (Llama-3.2-1b, Nova-micro) drop to 2%. This wide range in RIES (113.6–186.4) motivates our investigation of what factors predict resistance.

4.2. Size Predicts RI but Not PI

What determines whether a model resists retroactive interference? We tested two candidate predictors: model size (parameter count) and context window length.

Size correlates with RI resistance. Linear regression reveals that parameter count accounts for 49% of variance in RIES—our aggregate measure of accuracy maintained across all interference levels ($R^2 = 0.491$, $\beta = 0.70$, $p < 0.0001$; Figure 3). Models in the largest size tier ($\geq$671B parameters) achieve mean RIES of 171.2, compared to 142.8 for the smallest tier ($\leq$10B)—a 28-point advantage, with ANOVA confirming a significant monotonic size effect ($F(3, 35) = 11.83$, $p < 0.0001$).

Table 1. **RI Resistance Varies Widely.** Top 5 and bottom 5 models by RIES, showing accuracy at low (N=3) and high (N=300) interference.

Model	RIES	Acc@3	Acc@300
Top 5			
o1	186.4	100%	100%
o1-preview	186.4	100%	99%
GPT-5	186.4	100%	38%
o3	185.8	100%	100%
o3-mini	183.3	100%	68%
Bottom 5			
Llama-3.2-3b	113.6	0%	76%
Llama-3.2-1b	126.1	74%	2%
Nova-micro	126.1	95%	2%
GPT-4.1-nano	128.2	83%	7%
Llama-3.1-8b	128.3	59%	14%

Context length does not. Context window size shows no relationship with RIES ($R^2 = 0.003$, $p = 0.746$). A model with 32K context and 100B parameters outperforms a model with 1M context and 10B parameters. This dissociation suggests that interference resistance reflects representational capacity, not buffer size.

Critically, size does NOT predict PI resistance. When we apply the same regression to PIES (the equivalent metric for PI), model size explains only 6% of variance ($R^2 = 0.06$, $p = 0.14$, n.s.). This asymmetry—size predicting RI but not PI—is consistent with the two interference types involving different processes.

4.3. Reasoning Models: Better RI, Worse PI

Do architectural innovations beyond scale affect interference resistance? We compared chain-of-thought reasoning models (Wei et al., 2022; OpenAI, 2024) (o1, o3, o4-mini, DeepSeek-R1; n=6) against standard models (n=33).

Reasoning models show 18% higher RIES. Mean RIES for reasoning models is 178.0 (SD=12.6) versus 150.3 (SD=18.2) for non-reasoning models—an 18.4% advantage (independent $t(37) = 3.54$, $p = 0.001$, Cohen's $d = 1.76$). Five of six reasoning models rank in the top 10 for RI resistance.

But this advantage does not transfer to PI. The same reasoning models that excel at RI show no advantage—and often disadvantage—for PI. Most strikingly, o1 ranks #1 for RI (RIES = 186.4) but #35 for PI (PIES = 21.9), an 8.5× asymmetry within a single model.

4.4. Error Patterns and the Metacognitive Asymmetry

We analyzed 5,409 RI errors and 9,753 PI errors across all 39 models. Errors were classified algorithmically into five categories by comparing each incorrect response against the full stimulus sequence: retrieval failure (no value returned), same-key interference (wrong value from the correct category), cross-key interference (value from a different category), hallucination (value never presented), and partial match.

Table 2. **Error Distribution: RI vs PI.**

Error Type	RI (%)	PI (%)
Retrieval failure	**50.8**	42.5
Same-key intrusion	46.0	**56.1**
Cross-key intrusion	0.6	0.7
Hallucination	0.9	0.2
Partial match	1.8	0.7

The metacognitive signal is asymmetric across RI and PI. Using LLM-as-judge classification on retrieval failures, we find that 8.6% of PI retrieval failures involve explicit refusal (e.g., "I'm sorry, but I can't determine that from the provided text"), compared to 0% for RI. Translated to overall error rates: PI carries a weak uncertainty signal (∼4% of PI errors produce refusal), while RI is fully silent. This is a behavioral observation—we do not test the mechanism behind the asymmetry in this work—and extends the "mostly-know-what-they-know" pattern of Kadavath et al. (2022) to a specific in-context retrieval regime: partial self-knowledge for one failure mode, none for the other.

Error-type signatures differ (Table 2). RI failures are predominantly retrieval failures (50.8%)—models cannot access any value for the queried category. PI failures are predominantly same-key intrusions (56.1%)—models return earlier values from the correct category. Hallucination is rare in both conditions (<1%), indicating interference is a retrieval/selection problem, not a generation problem. RI failures thus appear "passive" (the initial encoding was overwritten or inaccessible), while PI failures appear "active" (earlier encodings intrude on attempts to access recent information).

Position bias confirms opposite retrieval dynamics. When same-key intrusions occur, RI intrusions show recency bias: errors come from middle-to-late positions (mean normalized position = 0.51). PI intrusions show primacy bias: errors come from early positions, with 14% returning the *first* value presented. This contrast is sharpest at minimal interference ($N = 3$): RI errors come from position 0.72 (late) versus PI errors from 0.12 (early)—direct behavioral evidence that the two interference types engage opposite retrieval dynamics.

5. Discussion

A Silent, Structured Epistemic Failure. The central finding is that LLMs fail silently under in-context interference:

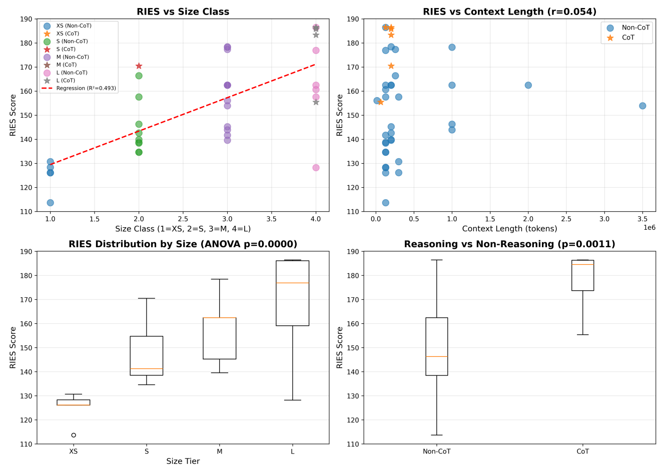

Figure 3. **Size Predicts RI Resistance, Not PI.** (A) Model size vs RIES shows strong positive relationship ($R^2 = 0.49$, $p < 0.0001$). (B) Context length vs RIES shows no relationship ($R^2 = 0.003$, n.s.). Size-PI relationship: $R^2 = 0.06$, n.s.—size does not predict PI resistance (see Figure 6 in Appendix).

more than 97% of errors produce no refusal or hallucination. This silence has internal structure—three dissociations by scale, by reasoning training, and by failure-type metacognition (§4.2–§4.4)—each pulling in a direction that a unified "memory capacity" account cannot explain. LLMs thus show partial self-knowledge about their memory failures, asymmetric across failure modes, extending the "mostly-know-what-they-know" pattern of Kadavath et al. (2022) to a specific in-context retrieval regime.

Is This Just "Lost in the Middle"? A natural alternative explanation is that our findings simply re-express known positional biases in transformer attention: models over-attend to initial tokens (Xiao et al., 2024) and under-attend to middle positions (Liu et al., 2024). This alternative account is partially correct—both effects point to a primacy bias—but does not predict our specific dissociations. The "lost in the middle" literature cannot explain why size predicts RI but not PI, why reasoning models invert the RI/PI trade-off, or why metacognitive signal differs across the two conditions. Positional bias is part of the story; the paper characterizes how that bias manifests as two distinct, structurally different failure modes under semantic interference.

Primacy Protection: A Mechanistic Sketch. Two hedged mechanistic accounts fit the data, though we do not test them directly. First, the *superposition* hypothesis (Elhage et al., 2022; Scherlis et al., 2022) offers one account for the size–RI correlation: smaller models are forced into greater representational overlap, blurring initial encodings under interference; larger models allocate more orthogonal representations. For PI, where attention must *select* among (distinctly represented) values rather than preserve them, additional parameters offer no advantage. Second, causal attention produces an inherent primacy bias: early tokens accumulate attention from all subsequent positions, while later tokens can only attend backward. Combined with the attention-sink effect (Xiao et al., 2024), this yields "primacy protection"—a structural feature, not a bug, that preserves distant context at the cost of recency, in contrast to the recency dominance typical of human immediate recall (Murdock, 1962; MacLeod, 2024).

6. Conclusion

We presented the first systematic comparison of retroactive and proactive interference in LLMs, characterizing a silent, structured epistemic failure. Three findings anchor the picture: (1) all 39 tested models show PI causing more degradation than RI ($d = 1.73$), inverting the typical human pattern of recency dominance in immediate recall; (2) model size predicts RI resistance ($R^2 = 0.49$) but not PI resistance ($R^2 = 0.06$, n.s.), and chain-of-thought reasoning improves RI at the cost of PI—scaling and architectural interventions address different failure modes; (3) the uncertainty signal is asymmetric: PI retrieval failures trigger refusal 8.6% of the time, RI failures 0%—partial self-knowledge for one failure mode, none for the other. Across both conditions, fewer than 1% of errors are hallucinations; more than 97% of wrong retrievals occur with no uncertainty signal of any kind.

Future Directions. Three directions extend this work. First, *ecologically valid benchmarks*: our synthetic category-value paradigm should be complemented by long-running narratives where facts evolve organically (Kočiský et al., 2018) and semi-structured evolving documents—medical case logs, legal case files, news timelines—where factual updates are critical and ground truth remains verifiable. Second, *mechanistic validation*: attention probing on open-weight models can directly test whether early tokens accumulate disproportionate attention, and analysis of training checkpoints (Biderman et al., 2023) can reveal whether the PI > RI asymmetry is architectural or learned. Third, *targeted mitigation*: once mechanisms are understood, principled interventions—recency-weighted attention, structured context ordering, positional debiasing (Liu et al., 2024)—become feasible.

Limitations

Our findings derive from a single experimental paradigm—the AB-AC paired-associate task. While this classic design enables controlled comparison between RI and PI, it is unclear whether the patterns generalize to other interference manipulations (e.g., varied retention intervals, intervening tasks) or alternative paradigms. The multi-mechanism reading of our dissociations, while consistent with our behavioral results, is not uniquely entailed by them: an alternative single-mechanism account—where RI and PI reflect two bottlenecks of the same attention process (representational dilution at encoding vs. winner-take-all competition at retrieval)—cannot be excluded by behavioral data alone and would require mechanistic probing to distinguish.

Additional limitations include: (1) parameter counts for closed-source models are estimates; (2) synthetic category-value stimuli may not capture naturalistic interference dynamics (see Future Directions); (3) our evidence is behavioral—mechanistic validation via attention probing is discussed in Future Directions; (4) all tested models are transformer-based, excluding state-space models and base versus fine-tuned comparisons; (5) English-only testing limits cross-linguistic generalization.

Ethical Considerations

This work evaluates memory interference in LLMs through controlled experiments that do not involve human subjects or personal data. All experiments used synthetic category-value pairs specifically designed for this research. Our findings on model-specific vulnerabilities (e.g., PI susceptibility in reasoning models) aim to inform responsible deployment rather than enable misuse. We note potential dual-use concerns: understanding interference patterns could theoretically be exploited to craft adversarial prompts that confuse model memory. However, we believe the defensive value—enabling practitioners to select appropriate models for interference-sensitive applications and anticipate failure modes—outweighs this risk. Model parameter counts for proprietary systems are estimates; we make no claims about internal architectures beyond publicly available information.

We encourage replication and extension of these findings across additional languages and model families to ensure generalizability.

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A. Experimental Setup

Stimuli. We used 46 semantic categories with 50–70 real-world meaningful values per category:

Arts: visual art, dance style, literary genre, painting medium. *Science:* chemical element, constellation, mathematical concept. *Nature:* bird species, flower species, tree species, weather phenomenon. *Food:* cheese variety, wine variety, pasta shape, culinary herb, fruit variety. *Culture:* ancient civilization, martial art, board game. *Technology:* programming language, telescope type, photography technique.

Example values for "fruit variety": mangosteen, bergamot, persimmon, mirabelle, kiwi, grape, etc.

Prompt Structure. Table 3 shows the three-phase RI prompt structure.

Phase	**Content**
Instruction	You will learn initial facts about categories, then see updates. Later, recall the INITIAL values you learned first.
Initial Learning	visual art: abstract expressionism tools: screwdriver ... *(46 categories)*
Updates	Following facts have been updated: visual art: baroque tools: hammer ... *($N \times 46$ updates, randomized order)*
Query	What was the INITIAL value of visual art?

Table 3. **RI Prompt Structure.** The prompt presents initial category–value pairs, followed by N rounds of updates, then queries a single category.

PI Prompt: Identical structure, except the query asks "What was the *LAST* value of visual art?" Both conditions use the same stimulus sequence—only the retrieval target differs.

Metric Definitions. Both RIES and PIES are computed as the area under the accuracy-vs-interference curve using trapezoidal integration with log-scaled x-axis. Let $\Delta_i = \log_{10}(L_{i+1}+1) - \log_{10}(L_i+1)$ for levels $L_i \in \{3, 10, 50, 100, 200, 300\}$:

$$\text{RIES} = \sum_{i=1}^{n-1} \frac{A_i^{\text{RI}} + A_{i+1}^{\text{RI}}}{2} \cdot \Delta_i \tag{2}$$

$$\text{PIES} = \sum_{i=1}^{n-1} \frac{A_i^{\text{PI}} + A_{i+1}^{\text{PI}}}{2} \cdot \Delta_i \quad (3)$$

where A_i^{RI} and A_i^{PI} are RI and PI accuracy at level L_i. The $+1$ offset accounts for the initial value presentation. Log-scaling ensures equal weighting across orders of magnitude. Higher scores indicate greater resistance.

B. Adaptive Stopping with Bootstrap CI

Each session draws a *fresh random sequence* from the value pool—different category-value assignments each time—ensuring true statistical independence across sessions despite deterministic (temperature $= 0$) generation. Sessions run until the 95% bootstrap confidence interval (CI_{95}; 10,000 resamples) narrows below 0.10 for two consecutive sessions, with a minimum of 5 sessions to prevent bootstrap degeneracy. Standard models have a hard cap of 50 sessions; expensive reasoning models (o-series, DeepSeek-R1) cap at 20. Sessions where the parser extracts fewer than 50% of categories—typically due to API errors or format mismatches rather than genuine model failures—are flagged as parse failures and excluded from the CI calculation.

Motivation. An initial pilot design used 3 fixed sessions with a single pre-built stimulus sequence per interference level. Post-hoc bootstrap analysis of those pilot runs revealed two problems. First, RI appeared artificially stable: 73.7% of model-level cells showed zero variance—but this reflected *identical prompts* yielding identical outputs at temperature 0, not genuine reproducibility. Second, PI showed systematic instability at moderate interference: 17.1% of cells had CI_{95} width ≥ 0.10, with several models exhibiting bimodal session accuracy (e.g., $[100\%, 0\%, 0\%]$ at $N = 3$) driven by sporadic API failures rather than true model variance. The adaptive design with fresh sequences per session eliminates both artifacts: genuine RI stability can be distinguished from prompt-repetition artifacts, and the bootstrap stopping rule ensures each reported mean has a well-characterized confidence bound.

C. Model Specifications

Table 7 lists all 39 models tested with complete data across all interference levels, sorted by parameter count (descending). RIES measures resistance to retroactive interference; PIES measures resistance to proactive interference. Architecture indicates Dense or Mixture-of-Experts (MoE); Reasoning indicates chain-of-thought models. **Note:** Models with optional extended thinking (e.g., Claude-4-opus, Gemini-2.5-pro) were tested with thinking *disabled*. Only models that use chain-of-thought by default (o-series, DeepSeek-R1) are classified as reasoning.

Models are grouped into four size tiers (Table 4) for decay curve analysis. The testbed spans seven model families (OpenAI, Anthropic, Google, Meta, Amazon, Alibaba, DeepSeek) covering parameter counts from 1B to 2,500B, both Dense and MoE architectures, and both standard and reasoning configurations.

Tier	Parameters	n	Examples
XS	≤10B	5	Llama-3.2-1b, Nova-micro
S	11–99B	10	Llama-3.2-90b, o4-mini
M	100–670B	13	Claude-4.5-opus, Llama-3.1-405b
L	≥671B	11	GPT-5, o1, DeepSeek-R1

Table 4. **Size Tier Definitions.** Parameter-based grouping used for decay curve analysis (Figures 5–6).

D. Error Type Taxonomy

All incorrect responses (5,409 total across 39 models and 6 interference levels) were classified into five mutually exclusive error types (Table 5). Examples assume the query asks for the *initial* value of "visual art" (correct: "abstract expressionism"), with N updates reassigning it to "baroque," "cubism," etc. Table 6 shows how the distribution shifts across interference levels.

Error Type	%	Definition & Example
Same-Key Interf.	46.0	Value from correct category, wrong position. *"baroque"*
Retrieval Failure	50.8	No value produced; refusal or empty. *"I'm not sure"*
Partial Match	1.8	Truncated or recombined. *"abstract express"*
Hallucination	0.9	Plausible but never presented. *"impressionism"*
Cross-Key Interf.	0.6	Value from a different category. *"screwdriver"*

Table 5. **Error Type Taxonomy.** Five mutually exclusive error categories (n=5,409 total errors). Sorted by frequency.

Error Type	Low *N=3,10*	Moderate *N=50,100*	High *N=200,300*
Same-Key Interf.	6.6%	32.7%	55.5%
Cross-Key Interf.	1.8%	0.4%	0.5%
Hallucination	5.0%	1.6%	0.1%
Retrieval Failure	78.4%	65.3%	42.3%
Partial Match	8.2%	0.0%	1.6%

Table 6. **Error Distribution by Interference Stage.** At low interference, errors are dominated by retrieval failures. As interference increases, same-key intrusions progressively dominate, indicating that competing memories become strong enough to override the target rather than simply blocking retrieval.

Model	B	Arch	Rea.	RIES	PIES	Model	B	Arch	Rea.	RIES	PIES
GPT-5	2500	MoE	N	186.4	99.2	Gemini-2.0-flash	100	MoE	N	162.5	101.6
GPT-5-mini	2500	MoE	N	162.5	109.1	Gemini-2.0-flash-lite	100	MoE	N	143.9	98.6
GPT-5-nano	2500	MoE	N	160.6	21.0	Llama-3.2-90b	90	Dense	N	134.6	87.1
o3	2000	MoE	Y	185.8	76.0	Nova-pro	70	Dense	N	157.5	63.7
o3-mini	2000	MoE	Y	183.3	75.6	Llama-3.1-70b	70	Dense	N	134.6	87.1
o1	1760	MoE	Y	186.4	22.0	Llama-3.3-70b	70	Dense	N	138.7	100.3
o1-preview	1760	MoE	Y	186.4	76.0	Gemini-2.5-flash-lite	40	MoE	N	146.3	91.3
GPT-4.1	1760	MoE	N	176.9	118.6	o4-mini	30	MoE	Y	170.5	22.0
GPT-4.1-mini	1760	MoE	N	157.6	116.9	Qwen-3-coder-30b	30	MoE	N	166.4	89.3
GPT-4.1-nano	1760	MoE	N	128.2	73.9	Claude-4.5-haiku	15	Dense	N	142.6	128.5
DeepSeek-R1	671	Dense	Y	155.4	55.6	Llama-3.2-11b	11	Dense	N	138.4	47.6
Llama-3.1-405b	405	Dense	N	141.7	57.2	Claude-3.5-haiku	10	Dense	N	139.8	126.4
Llama-4-maverick	400	MoE	N	178.3	67.9	Llama-3.1-8b	8	Dense	N	128.3	57.1
Claude-4.5-opus	300	Dense	N	178.4	170.3	Nova-lite	7	Dense	N	130.7	53.2
Gemini-2.5-pro	300	MoE	N	162.5	123.6	Llama-3.2-3b	3	Dense	N	113.6	16.8
Claude-4.5-sonnet	250	Dense	N	162.5	124.2	Nova-micro	1	Dense	N	126.1	76.8
Qwen-3-VL-235b	235	MoE	N	177.3	120.2	Llama-3.2-1b	1	Dense	N	126.1	1.0
Claude-4-opus	200	Dense	N	139.6	131.0						
Claude-4-sonnet	200	Dense	N	162.5	137.8						
Claude-3-opus	175	Dense	N	145.2	131.5	*B = parameters in billions*					
GPT-3.5-turbo	175	MoE	N	156.1	120.0	*Rea. = reasoning model (Y/N)*					
Llama-4-scout	109	MoE	N	153.9	40.2						

Table 7. **Complete Model Specifications.** All 39 models with RIES and PIES scores. Models sorted by parameter count (descending). See Section C for details.

E. Detailed Results by Model and Condition

E.1. Top and Bottom Performers

Table 8 shows RI and PI accuracy at each interference level for the five highest- and lowest-scoring models. RI top performers are reasoning models maintaining near-ceiling accuracy; PI top performers are exclusively Claude models. Notably, o1 ranks #1 for RI but appears in the PI bottom 5 at 0%—the starkest RI/PI dissociation.

E.2. Reasoning vs Non-Reasoning Models

The six reasoning models (o1, o1-preview, o3, o3-mini, o4-mini, DeepSeek-R1) exhibit the most extreme RI–PI dissociation. Table 9 compares mean accuracy by interference level; Table 10 provides the per-model breakdown.

Why 0% PI accuracy? The 0% scores do *not* reflect refusals or empty responses. These models respond confidently with all 46 category values—but consistently return values from *earlier* positions rather than the most recent ones. The chain-of-thought process appears to retain earlier encodings so strongly that recent updates become inaccessible, even when the prompt explicitly requests the "LAST value." This is primacy intrusion, not retrieval failure. Table 11 shows a representative example.

F. Supplementary Analyses

Decay Function Fits Across Architectures. Unlike PI, which follows a universal log-linear decline across all models (Wang and Sun, 2025), RI decay shows architectural diversity. Fitting decay functions to each model's accuracy curve reveals: 35% follow exponential decay (rapid early forgetting), 30% polynomial (gradual degradation), 26% log-quadratic (initial resistance then collapse), and 9% power-law (O-series reasoning models).

Correlation Between RI and PI Resistance. If RI and PI reflect a unified "memory capacity," models good at one should be good at the other. Across 39 models, Pearson correlation yields $r = 0.21$ (95% CI $= [-0.11, 0.49]$, $p = 0.20$; Figure 4). With only 39 models the CI crosses zero, so we cannot cleanly separate "weakly correlated" from "uncorrelated." What we can say: there is no evidence of the strong correlation a unified-capacity account would predict. This absence, combined with the size and reasoning dissociations reported in the main body, is consistent with RI and PI drawing on separate processes, though our sample is underpowered to detect moderate correlations.

Asymmetry Magnitude Across Model Families. While all 39 models show PI $>$ RI, the magnitude varies across families: o-series reasoning models show the most extreme asymmetry (o1: $8.5\times$, RIES = 186.4 vs. PIES = 21.9). Claude models show minimal asymmetry (Claude-4.5-opus: $1.05\times$, RIES = 178.4 vs. PIES = 170.3; seven of the top

Model	RI Accuracy (%) 3	10	50	100	200	300	Model	PI Accuracy (%) 3	10	50	100	200	300
Top 5 (highest RIES)							*Top 5 (highest PIES)*						
o1	100	97.8	98.6	100	100	100	Claude-4.5-opus	100	97.8	91.3	80.4	84.8	71.7
o1-preview	100	97.8	100	100	98.6	99.3	Claude-4-sonnet	100	97.8	93.5	34.8	23.9	8.7
GPT-5	100	97.8	100	100	98.6	38.4	Claude-3-opus	100	84.8	95.7	32.6	28.3	2.2
o3	100	97.8	100	99.3	99.3	100	Claude-4-opus	100	91.3	91.3	30.4	21.7	4.3
o3-mini	100	97.8	100	65.9	43.5	68.1	Claude-4.5-haiku	100	93.5	95.7	19.6	10.9	4.3
Bottom 5 (lowest RIES)							*Bottom 5 (lowest PIES)*						
Llama-3.1-8b	58.7	97.8	89.1	43.5	16.7	13.8	o1	100	0	0	0	0	0
GPT-4.1-nano	82.6	71.0	87.7	56.5	23.9	6.5	o4-mini	100	0	0	0	0	0
Nova-micro	94.9	89.1	69.6	35.5	15.2	2.2	GPT-5-nano	95.7	0	0	0	0	0
Llama-3.2-1b	73.9	97.8	81.9	56.5	0	2.2	Llama-3.2-3b	73.9	0	0	0	2.2	0
Llama-3.2-3b	0	89.1	97.8	21.7	17.4	76.1	Llama-3.2-1b	2.2	0	0	0	2.2	0

Table 8. **RI and PI Accuracy (%) at Each Interference Level.** Top 5 and bottom 5 models by RIES (left) and PIES (right).

	3	10	50	100	200	300
RI Accuracy (%)						
Reasoning	100	97.8	99.8	94.1	82.0	73.4
Non-reasoning	92.8	92.9	95.3	69.1	38.9	15.7
Δ	**+7.2**	**+4.9**	**+4.4**	**+25.0**	**+43.1**	**+57.7**
PI Accuracy (%)						
Reasoning	99.6	59.1	0	0	0	0
Non-reasoning	93.5	69.4	53.6	10.9	9.9	3.4
Δ	+6.2	-10.3	$\mathbf{-53.6}$	-10.9	-9.9	-3.4

Table 9. **Reasoning vs Non-Reasoning Models.** Mean accuracy by interference level. Reasoning models show a widening RI advantage (+57.7% at N=300) but catastrophic PI failure (0% from N=50 onward).

Model	RI Accuracy (%) 3	10	50	100	200	300	PI Accuracy (%) 3	10	50	100	200	300
o1	100	97.8	98.6	100	100	100	100	0	0	0	0	0
o1-preview	100	97.8	100	100	98.6	99.3	100	97.8	0	0	0	0
o3	100	97.8	100	99.3	99.3	100	100	97.8	0	0	0	0
o3-mini	100	97.8	100	65.9	43.5	68.1	97.8	97.8	0	0	0	0
o4-mini	100	97.8	100	100	55.1	5.8	100	0	0	0	0	0
DeepSeek-R1	100	97.8	100	99.3	95.7	67.4	100	60.9	0	0	0	0

Table 10. **Reasoning Models: RI vs PI Accuracy at Each Interference Level.** Most models maintain high RI accuracy through N=100, though o3-mini and o4-mini degrade earlier. All six collapse to 0% PI by N=50, with o1 and o4-mini failing at N=10—the earliest collapse point.

Field	Value
Model	o1
Condition	PI, N=300
Category	visual art
Expected (last)	land art (update #300)
Returned	“neo pop 176” (update #176)
The model responds confidently with position-indexed values: “visual art → neo pop 176”—returning update #176 out of 300, not the last. Despite explicitly tracking updates (“I have carefully tracked every update...”), it retrieves mid-sequence values. Accuracy: 0/46 (0%). The model’s reasoning process cannot overcome primacy bias.	

Table 11. **Example PI Failure: o1 at N=300.** The model returns earlier update values (with position indices) instead of the most recent ones, demonstrating primacy intrusion despite explicit chain-of-thought reasoning.

Field	Value
Model	Claude-4.5-haiku
Condition	RI, N=300
Category	literary genre
Expected (initial)	memoir
Returned	hard sci fi
The model returns "hard sci fi" (a later update value) instead of the initial value "memoir." At N=300, with 300 intervening updates per category, the initial encoding has been overwritten by subsequent information. Accuracy: 4/46 (8.7%). Unlike PI failures where earlier values intrude, RI failures show later values displacing the original—recency overwrites primacy.	

Table 12. **Example RI Failure: Claude-4.5-haiku at N=300.** The model returns a later update value instead of the initial one, demonstrating how retroactive interference overwrites early encodings with recent information.

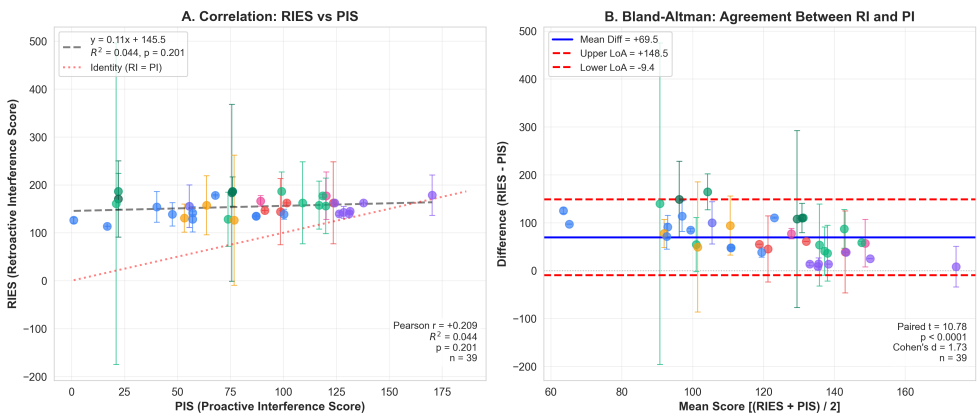

Figure 4. **RI and PI: Weak Correlation with Systematic Asymmetry.** (A) Scatter plot: $r = 0.21$ (95% CI $= [-0.11, 0.49]$); CI crosses zero. All points above the diagonal indicate universal RIES $>$ PIES. (B) Bland-Altman plot: systematic bias of $+69.5$ points (95% limits: -9.4 to $+148.5$).

10 PI performers are Claude). GPT-5 occupies the middle (1.9×). The 8-fold range indicates that the universal PI $>$ RI pattern admits substantial architectural variation: chain-of-thought training appears to trade PI resistance for RI, whereas Claude's training recipe shows balanced performance on both.

G. Additional Figures

This section presents supplementary visualizations that complement the tabular results above. Figures 5–6 show RI and PI decay curves stratified by model size tier. Figure 7 contrasts reasoning and non-reasoning models. Figures 8–9 provide detailed error pattern visualizations.

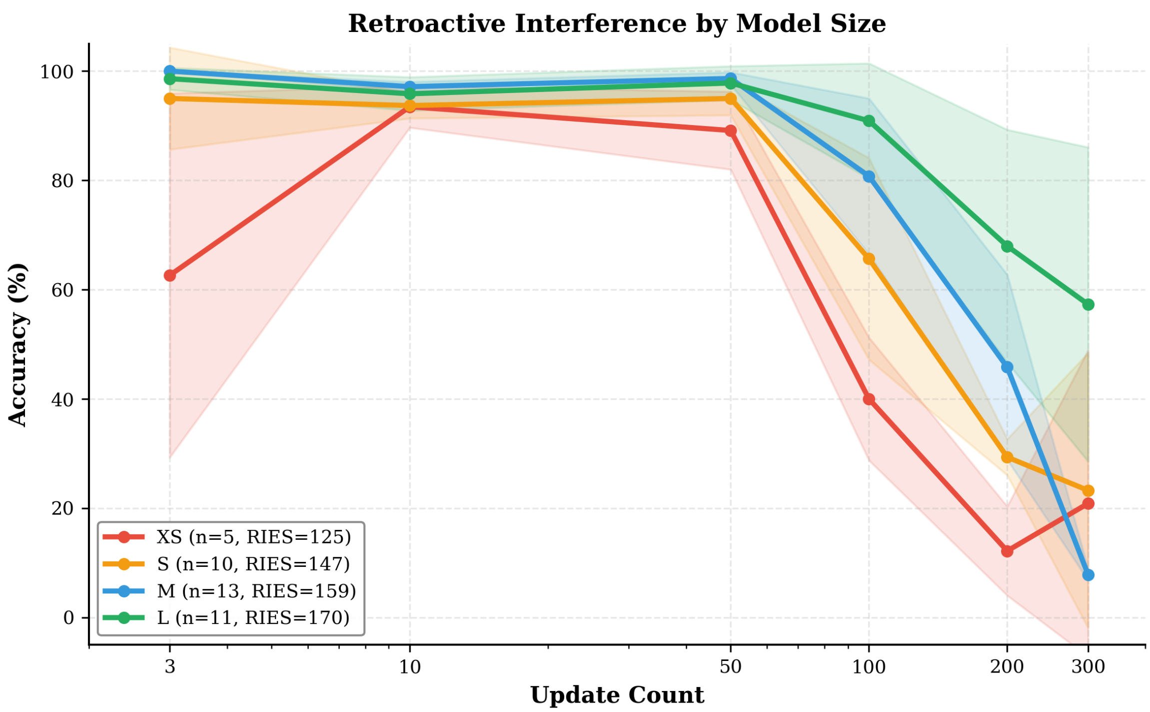

Figure 5. **RI Decay Curves by Model Size Tier.** Larger models (>100B parameters) maintain higher accuracy across all interference levels. The decay pattern shows that size provides consistent protection against retroactive interference, with the largest models maintaining >50% accuracy even at N=300.

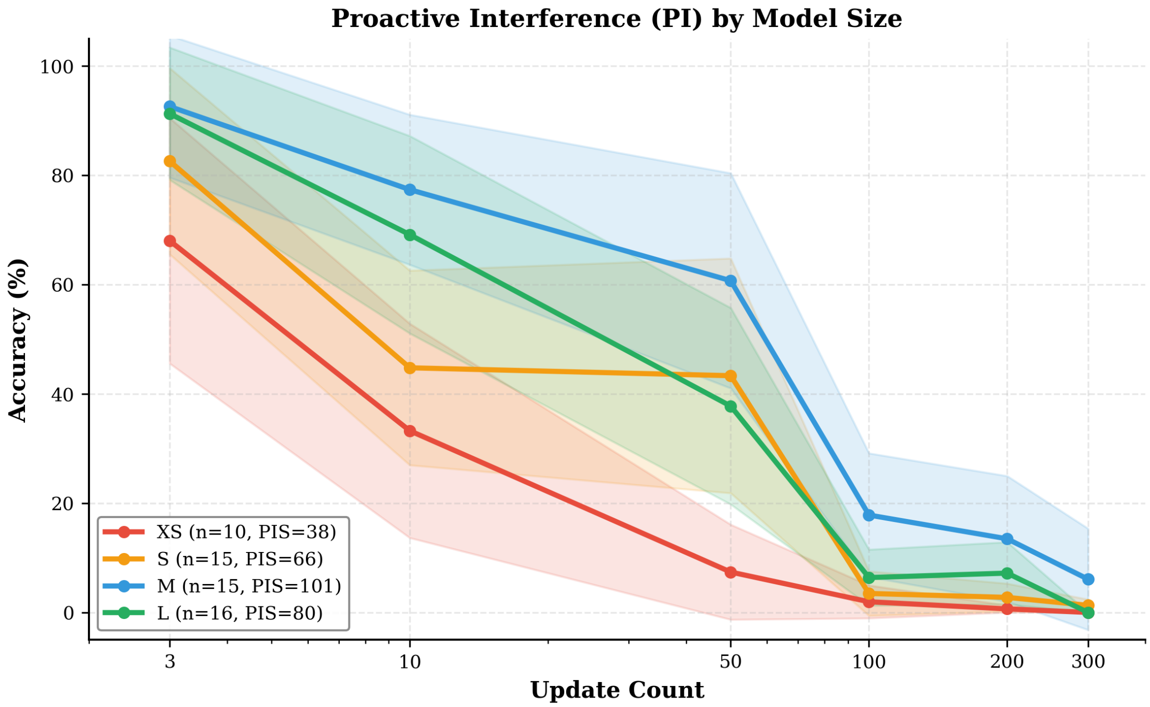

Figure 6. **PI Decay Curves by Model Size Tier.** Unlike RI (Figure 5), PI decay shows no consistent relationship with model size ($R^2 = 0.06$, n.s.). Medium-sized models (M tier, blue) actually outperform large models (L tier, green) at high interference levels. This lack of size-dependence, contrasting with RI's strong size correlation ($R^2 = 0.49$), provides evidence that PI is architecture-constrained rather than capacity-limited.

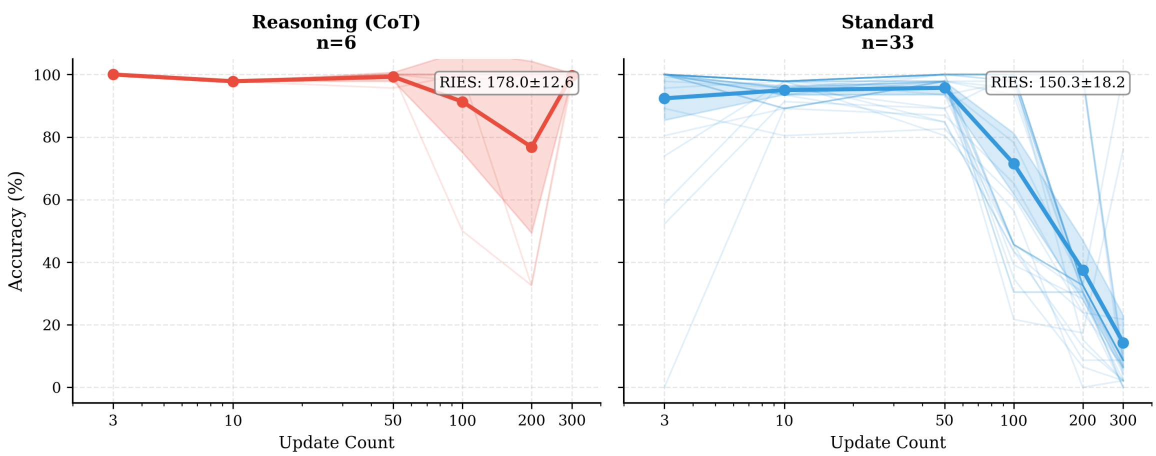

Figure 7. **RI Decay Curves: Reasoning vs Non-Reasoning Models.** Chain-of-thought reasoning models (solid lines) show markedly slower decay compared to standard models (dashed lines). Extended inference-time computation appears to enhance retention of initial encodings, with reasoning models maintaining higher accuracy at all interference levels.

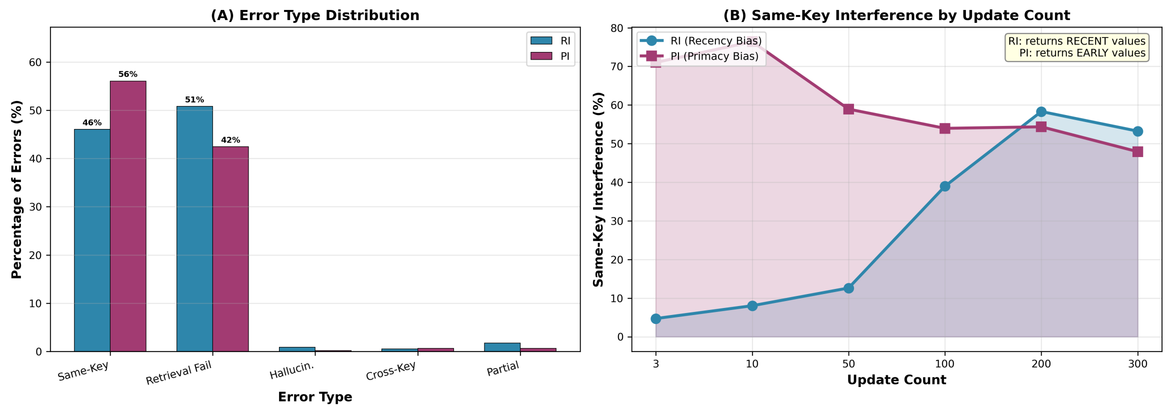

Figure 8. **Detailed Error Pattern Analysis.** Distribution of error types across RI and PI conditions. RI errors are predominantly retrieval failures (50.8%), while PI errors show primacy intrusion (56.1% same-key interference). Both conditions show minimal hallucination (<1%), confirming that models confuse positions rather than fabricate values.

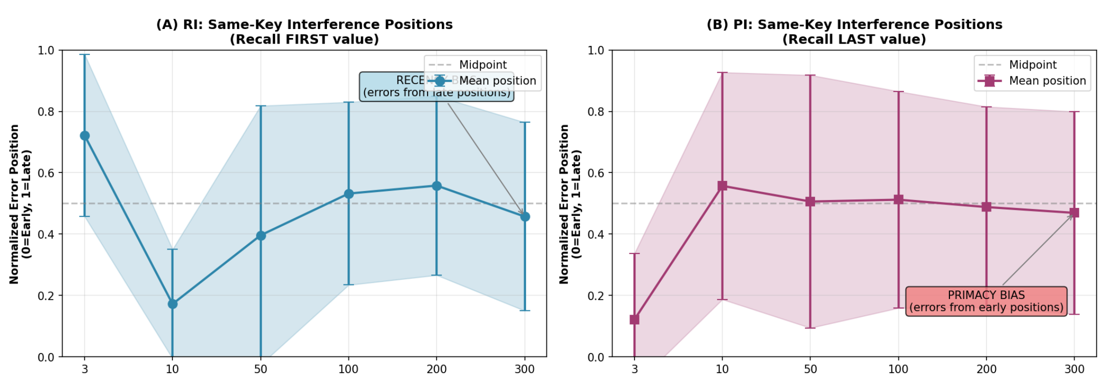

Figure 9. **Position-Based Error Analysis.** When same-key intrusions occur, RI errors come from middle-to-late positions (recency bias), while PI errors come from early positions with 14% returning the first value (primacy bias). This opposing directionality provides mechanistic evidence for distinct retrieval dynamics.

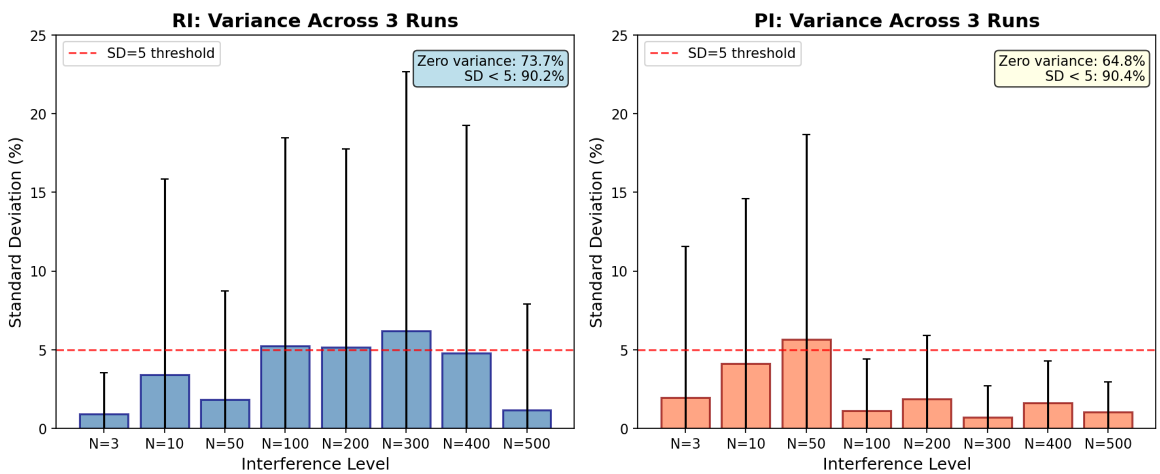

Figure 10. **Pilot Run Variance Distribution (Motivating the Adaptive Design).** Standard deviation of accuracy across 3 pilot runs using fixed stimulus sequences, aggregated by interference level for RI (blue) and PI (orange). While apparent variance was low (RI: 73.7% exact zero, 90.2% SD < 5; PI: 64.8% exact zero, 90.4% SD < 5), post-hoc bootstrap analysis (see Appendix B) revealed that this stability was largely an artifact of identical prompts at temperature 0—not genuine reproducibility. PI cells with bimodal outcomes ($[100\%, 0\%, 0\%]$) reflected sporadic API failures. These findings motivated the adaptive design with fresh random sequences per session.